BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Linear Search

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

1. Why must a trace table follow executed assignments in order?

2. What does lossy compression remove?
   A) Only the file's name
   B) All binary digits
   C) Only duplicate file copies
   D) Some original information

3. "Changing a variable name automatically fixes every logic error." Correct this. Explain why.

4. A program is compressed before being sent. "Lossless compression removes source-code characters." Correct this. Why does it matter here?

Today's objectives

Explain the steps and required conditions for linear search. Apply the algorithm or method to supplied data.

Key vocabulary - Read, cover, recall

Linear search
Checks one element after another.

Target
Value being searched for.

Index
Zero-based position of an array element.

Comparison
Test of current element against target.

BLACK PEN: Write one definition in your own words.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Learn and practise

New Learning - Key ideas

Works without sorting

Linear search inspects items in sequence, so the list does not need to be sorted.

Check each element

Compare the target with the item at index 0, then increase the index if there is no match.

Report the result

Stop on the first match and return/report its index, or report not found after the last item.

Count comparisons

For n items, a search may check up to n items; finding an early match can require fewer.

I Do - Worked model

Trace linear search for target 14 in [6,12,9,14,20,3,11]. Use zero-based indexes. [4 marks]

- At index 0
- 6 · 14, so continue to the next element.
- Index 1 = 12 and index 2 = 9 are not the target.
- Index 3 = 14; found at index 3 after four comparisons. [4]

We Do - Guided practice

1. Find 15 in [4,8,12,15,19]. Give its index and comparisons. [4]

2. Find 7 in [9,3,8,2]. Describe the outcome. [3]

3. Explain why a linear search works on an unsorted list. [3]

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Complete the comparison that searches for the PC register.

```
array registers = ["MAR","MDR","PC","ACC"]
for i = 0 to 3
if ... then
print(i)
endif
next i
```

You Do - Independent exam questions

Q2. A linear search looks for 11 in [4,11,6,18,2]. State the index and number of comparisons. [3]

Q3. Explain why linear search can find a target in an unsorted list. [2]

Q4. A linear search fails to find 5 in [1,4,8,9]. How many elements are checked and why? [3]

Exit Ticket - True or False?

Binary search can safely discard half only when data is sorted.

TRUE [ ] **FALSE [ ]**

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: Tick each correct point; correct mistakes.

Do Now - Model answers

1. Each variable value depends on the state produced by earlier executed instructions.

2. D - Some original information

3. Clear names show what values mean and help people read and change code.

4. Lossless compression restores every character exactly. Losing characters could break the program. A thumbnail can lose some detail; source code cannot.

I Do - Explain the model

At index 0

6 · 14, so continue to the next element.

Index 1 = 12 and index 2 = 9 are not the target.

Index 3 = 14; found at index 3 after four comparisons. [4]

We Do - Model answers

1. Compare 4,8,12,15 [2]; target found at index 3 [1], after four checks [1].

2. All four values are checked [2]; target not found [1].

3. It examines every element in sequence [1], does not rely on ordering [1] and stops at a match or the end [1].

Code First - Answer

```
array registers = ["MAR","MDR","PC","ACC"]
for i = 0 to 3
if registers[i] == "PC" then
print(i)
endif
next i
```

Expected output: 2

You Do - Self-mark

Q2: Index 1 [1], following checks of 4 then 11 [1], total 2 comparisons [1].

Q3: Checks elements in order [1] without requiring sorting [1].

Q4: Four elements [1], because each item is compared [1] before concluding target absent [1].

GREEN PEN: Choose the hardest answer. Correct it on lined paper.